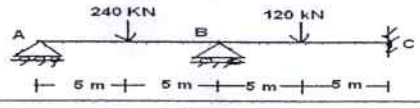
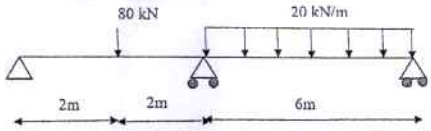
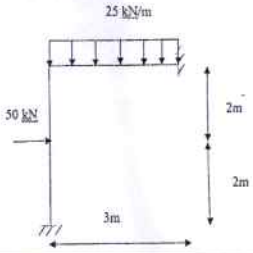
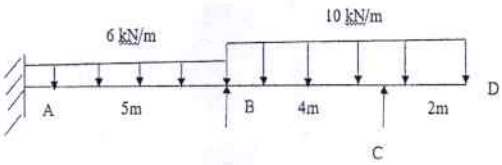
Roll No: | | | | | | | | |

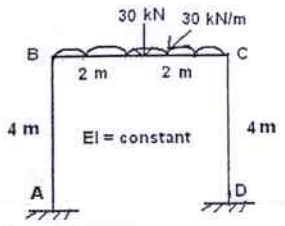
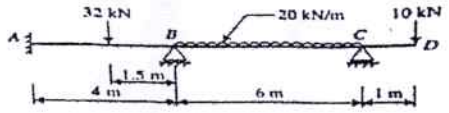
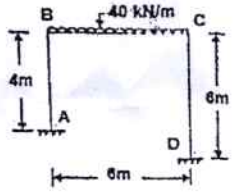
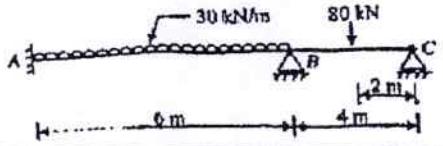
MID SEMESTER EXAMINATION- OCTOBER 2025

Name of the Course: M. Tech (Working Professional)
Name of the Paper: Matrix Method for Structural Analysis
Time: 1½ Hours

Semester : 1st semester
Paper Code: MSE 104
Maximum Marks: 50

Note: i) All questions are compulsory
ii) Each question carries 10 Marks

Q.1		(10 Marks)	
	a)	Analyze the following beam by stiffness matrix method. 	(CO 1)
		OR	
	b)	Answer the following (a) Define static indeterminacy. (b) Why the stiffness matrix method is also called equilibrium method or displacement method? (c) Define flexibility of structure. (d) Enlist the properties of the stiffness matrix. (e) Write down the equation of element stiffness matrix as applied to 2D plane element.	(CO 1)
Q.2		(10 Marks)	
	a)	Analyze the following beam by stiffness matrix method. 	(CO 1)
		OR	
	b)	Analyze the following portal frame by stiffness matrix method 	(CO 2)
Q.3		(10 Marks)	
	a)	Analyze the following beam by stiffness matrix method. 	(CO 1)

		OR	
	b)	Analyse the following frame by stiffness matrix method. 	(CO 2)
Q.4		(10 Marks)	
	a)	Analyse the following beam by stiffness matrix method 	(CO1)
		OR	
	b)	Analyse the following frame by stiffness matrix method. 	(CO 2)
Q.5		(10 Marks)	
	a)	Analyse the following beam by stiffness matrix method. 	(CO 1)
		OR	
	b)	Answer the following questions. (a) What is the relation between flexibility and stiffness matrix? (b) What do you mean by degree of freedom and kinematic indeterminacy? Give suitable example for them. (c) Is it possible to develop the flexibility matrix for an unstable structure? (d) What do you mean by global stiffness matrix? (e) What is the difference between flexibility method and stiffness method.	(CO 2)